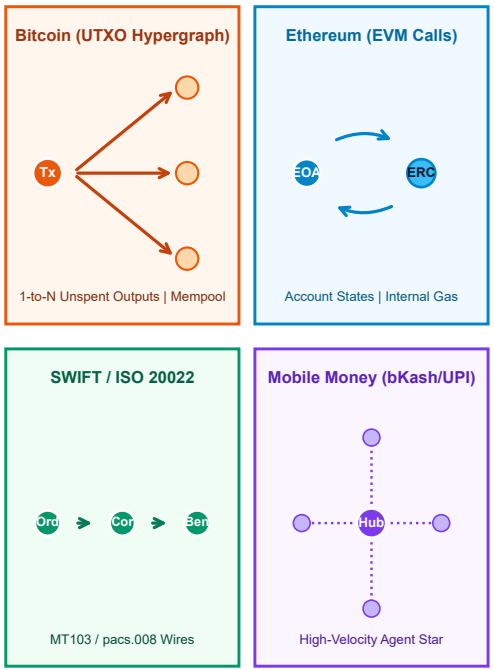
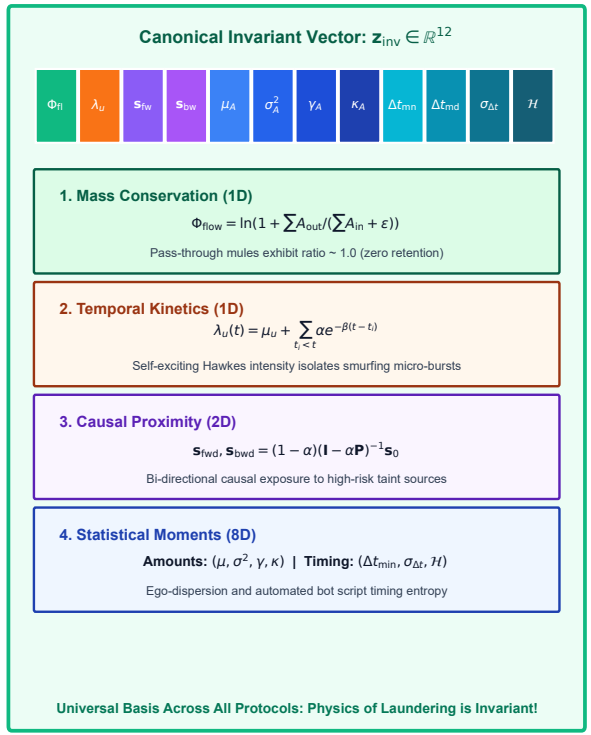
Universal 12-Dimensional Canonical Invariant Space: Cross-Ledger Zero-Shot Generalization

Deterministic Invariant Normalization: Heterogeneous Protocols -> Universal Manifold R^12 -> Frozen GNN Backbone

(a) Incompatible Protocol Topologies (Feature Dimensions: $d_1 \neq d_2 \neq d_3 \neq d_4$)



(b) Universal 12-D Invariant Prism $\mathbf{z}_{inv} \in \mathbb{R}^{12}$ (Mathematical Physics Normalization)



(c) Zero-Shot Cross-Ledger Transfer Matrix (Frozen C-STGB Backbone $\eta = 0$, Zero Fine-Tuning)

Target Ledger (Zero-Shot Deployment)

Source (Train)	BTC	ETH	Fiat	Credit
BTC	91.4% (Tuned)	82.4% (Zero-Shot)	76.8% (Zero-Shot)	79.2% (Zero-Shot)
ETH	78.5% (Zero-Shot)	97.4% (Tuned)	72.1% (Zero-Shot)	85.1% (Zero-Shot)
Fiat	71.4% (Zero-Shot)	68.2% (Zero-Shot)	93.7% (Tuned)	39.4% (Zero-Shot)
Credit	80.6% (Zero-Shot)	75.3% (Zero-Shot)	74.5% (Zero-Shot)	97.9% (Tuned)

Table 7.12 Empirical Zero-Shot Macro Transfer F1:

HGT: 19.51% *vs XGBoost 54.06% (Schema Drop)*

C-STGB Zero-Shot: 69.98% Macro F1 [+50.5 pp over HGT]